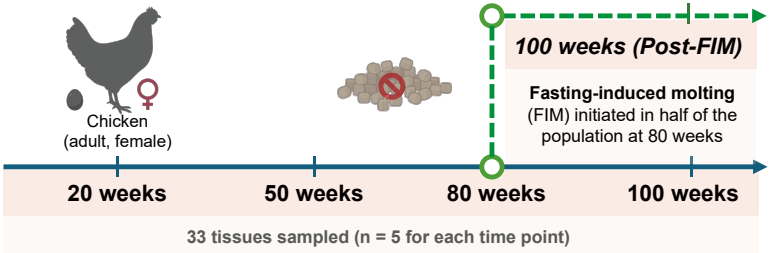
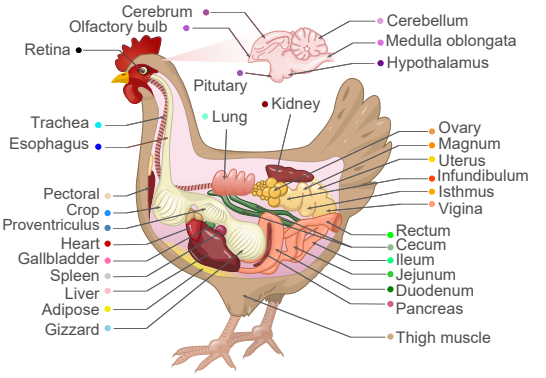
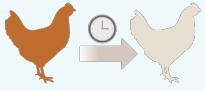
a Data generation



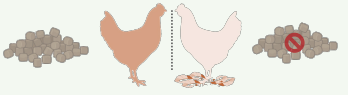


1 Natural Aging Model

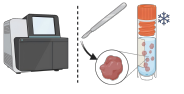


Natural aging of laying hens from 20 weeks to 100 weeks

2 FIM intervention Model



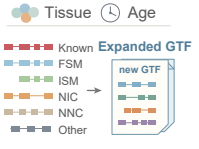
Comparison of age-matched control and FIM-treated hens at 100 weeks



825 Samples 4 Time points
33 Tissues FIM treatment
ONT long-read RNA sequencing

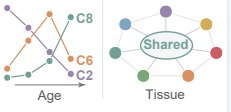
b Chicken productive aging atlas

Isoform discovery



Multidimensional chicken transcript atlas

Aging functional network



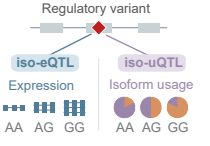
Shared and tissue-specific aging programs

Splicing dynamics



Aging-associated splicing dynamics

Transcript-level QTLs



Genetic regulation of transcript expression and usage

FIM restoration

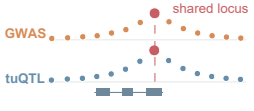


Selective restoration of aging-associated programs

c Applications and data sharing

QTL–trait colocalization

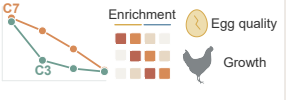
Transcript-level molecular phenotypes



Resolves trait-linked regulatory isoforms

Aging-program enrichment

167 chicken complex traits



Links aging trajectories to complex traits of chicken

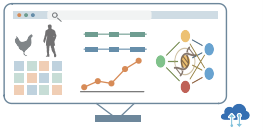
Cross-species comparison

805 complex traits GWAS of human



Cross-species heritability enrichment

Open atlas portal



Enables exploration and data reuse